

Long Division Masterclass

3-Digit by 1-Digit Numbers with Remainders

1. What is Long Division with Remainders?

Long division helps us divide large numbers by smaller ones. [cite: 14] Sometimes the numbers don't divide evenly, and we get leftovers called **remainders**. [cite: 15] This happens when the divisor doesn't fit perfectly into the dividend! [cite: 16]

The Candy Example

Imagine sharing **10** candies among **3** friends. Each friend gets **3** candies, and **1** candy is left over! That extra candy is the remainder. [cite: 20]

2. Step-by-Step Guide

Let's look at how to solve **$456 \div 7$** : [cite: 24]

- ✓ **Divide:** **$4 \div 7 = 0$** , so we try **$45 \div 7 = 6$** . [cite: 26]
- ✓ **Multiply & Subtract:** **$6 \times 7 = 42$** . Then, **$45 - 42 = 3$** . [cite: 27]
- ✓ **Bring Down:** Bring down the **6** to make **36**. [cite: 28]
- ✓ **Divide Again:** **$36 \div 7 = 5$** . [cite: 29]

- ✓ **Final Subtract:** $5 \times 7 = 35$. Then, $36 - 35 = 1$. [cite: 29, 30]
- ✓ **The Result:** Your answer is **65** with a Remainder of **1**! [cite: 31]

3. Real-World Uses

Why do we learn this? Division with remainders is useful for: [cite: 60]

- Sharing sweets among friends fairly [cite: 61]
- Packing items into boxes evenly [cite: 61]
- Splitting money and saving the leftovers [cite: 61]

4. Practice Problems

Try these on your own to become a division star! [cite: 53]

1. $456 \div 7 = ?$ [cite: 54]
2. $789 \div 4 = ?$ [cite: 55]
3. $523 \div 6 = ?$ [cite: 56]

Top Tips for Success

- Break the problem into small steps. [cite: 64]
- Check your multiplication carefully. [cite: 64]
- Use estimation to guess the answer first! [cite: 65]

Keep Practicing and You'll Be a Division Star! [cite: 67]

References: Long Division with Remainders for Kids (3).pdf